SUNY Old Westbury 2026-2027 Undergraduate Catalog

Biological Sciences, B.S.

HEGIS Number: 0401 Curriculum Code: U0313

Requirements for a Bachelor of Science (B.S.) Degree in Biological Sciences

(Total Credits Required 120)

Liberal Education Curriculum

Refer to the Liberal Education Curriculum Bulletin for specific requirements.

Departmental Requirements: 66 Credits

A grade of D may be accepted toward graduation in two of the required courses in Chemistry/Physics or Math.

Basic Biology Requirement: 8 Credits (2 courses)

- BS 2400 - Basic Biosciences I Credits: 3
- BS 2401 - Basic Biosciences I Lab Credits: 1

- BS 2410 - Basic Biosciences II Credits: 3
- BS 2411 - Basic Biosciences II Lab Credits: 1

Senior Seminar Requirement: 4 Credits (1 course)

- BS 5400 - Senior Seminar Credits: 4

Chemistry and Physics Requirement: 26 Credits (6 courses)

- CP 2120 - Principles of Chemistry I Credits: 3
- CP 2121 - Principles of Chemistry Lab I Credits: 1 &
- CP 2130 - Principles of Chemistry II Credits: 3
- CP 2131 - Principles of Chemistry Lab II Credits: 1

- CP 3300 - Organic Chemistry I Credits: 3
- CP 3302 - Organic Chemistry Lab I Credits: 2 &
- CP 3310 - Organic Chemistry II Credits: 3
- CP 3312 - Organic Chemistry Lab II Credits: 2

- PX 2000 - Physics 1, Algebra-based Credits: 3
- PX 2001 - Physics 1, Algebra Based Lab Credits: 1 &
- PX 2100 - Physics 2, Algebra-based Credits: 3
- PX 2101 - Physics 2, Algebra-based Lab Credits: 1

OR

- PX 2200 - Physics 1, Calculus-based Credits: 3
- PX 2201 - Physics 1, Calculus-based lab Credits: 1 &
- PX 2300 - Physics 2, Calculus-based Credits: 3
- PX 2301 - Physics 2, Calculus-based Lab Credits: 1

Mathematics

A grade of D may be accepted toward graduation in two of the required courses in Chemistry/Physics or Math.

- MA 2310 - Calculus & Analytic Geometry I Credits: 4

Upper Division Biology Requirement: 24 Credits (6 courses)

1. At least one course chosen from each of the following areas:

a. Cellular and Molecular Biology

- BS 4400 - Cell Biology Credits: 4
- BS 4460 - Genetics Credits: 4
- BS 4560 - Molecular Biology Credits: 4
- BS 3910 - Intro to Bioinformatics Credits: 4

b. Organismal Biology

- BS 3400 - Vertebrate Physiology Credits: 4
- BS 3520 - Comparative Anatomy Credits: 4
- BS 4430 - Developmental Biology Credits: 4

- BS 4410 - Histology Credits: 4

c. Ecology and Evolutionary Biology

- BS 4440 - Evolution Credits: 4
- BS 4470 - Ecology Credits: 4
- BS 4474 - Microbial Ecology Credits: 4
- BS 3530 - Comparative Genomics Credits: 4

2. Three courses chosen from the list of Biology electives (at least one must have a lab component)

Biology Electives

- BS 3400 - Vertebrate Physiology Credits: 4
- BS 3450 - Plant Biology Credits: 4
- BS 3520 - Comparative Anatomy Credits: 4
- BS 3530 - Comparative Genomics Credits: 4
- BS 3810 - Biological Aspects of Aging Credits: 4
- BS 3910 - Intro to Bioinformatics Credits: 4
- BS 4400 - Cell Biology Credits: 4
- BS 4410 - Histology Credits: 4
- BS 4420 - Microbiology Credits: 4
- BS 4430 - Developmental Biology Credits: 4
- BS 4440 - Evolution Credits: 4
- BS 4565 - Pharmacology Credits: 4
- BS 4460 - Genetics Credits: 4
- BS 4461 - Immunology Credits: 4
- BS 4470 - Ecology Credits: 4
- BS 4474 - Microbial Ecology Credits: 4
- BS 4480 - Animal Behavior Credits: 4
- BS 4500 - Neurobiology Credits: 4
- BS 4550 - Cancer Cell Biology Credits: 4
- BS 4560 - Molecular Biology Credits: 4
- BS 5590 - Advanced Research Credits: 4
- CP 4490 - Biochemistry for Life Sciences Credits: 3

- CP 4510 - Biochemistry I Credits: 3

3. Completion of BS 2400/BS 2401-BS 2410/BS 2411 Basic Biological Sciences I and II and Labs

With grades of C or better are prerequisites for upperdivision biology courses. A grade of C or better is required in each biology course in order to be credited for graduation.

- BS 2400 - Basic Biosciences I Credits: 3
- BS 2401 - Basic Biosciences I Lab Credits: 1
- BS 2410 - Basic Biosciences II Credits: 3
- BS 2411 - Basic Biosciences II Lab Credits: 1

4.A student may not repeat any Biology course more than one time.

General Electives

In consultation with Academic Advisor

Environmental Studies and Ecology

For students interested in the environment and ecology, the Biological Sciences Department offers a minor in Environmental Studies. Students should meet with an advisor for guidance in selecting electives and preparing their senior project.

A B.S. degree with this emphasis will prepare a student for advanced laboratory research in environmental biology while a B.A. degree with this emphasis is designed for students more interested in administering environmental programs. Direct participation in environmental agencies and organizations on Long Island or independent laboratory research on an environmental problem is on important aspect of this emphasis.